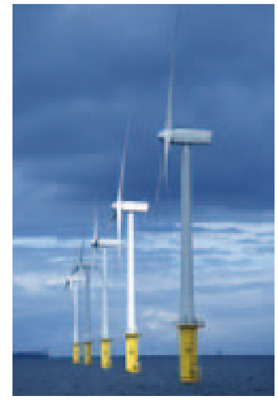
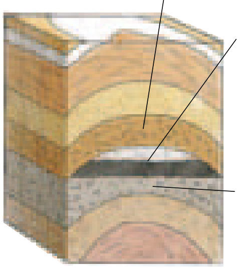
Oil and gas exploration

VALUABLE RESERVOIRS OF OIL AND GAS lie hidden in rocks beneath the seabed. Oil and gas are tapped by drilling down through the rocks, but first geologists must know where to drill. Only certain kinds of rocks hold oil and gas, but must be in shallow enough water to be reached by drilling. Geologists find the reservoirs by sending shock waves through the seabed and using the returning signals to distinguish between the rock layers. Temporary rigs are set up to pinpoint a source to see if the oil is the right quality and quantity. To extract oil or gas, the rig is replaced by a more permanent oil platform, which is firmly anchored to the seabed. Oil can be piped ashore or processed and stored on floating vessels (FPSOs) until it is offloaded onto tankers. When reservoirs dry up, new sources have to be found as there is a great demand for energy, but the Earth's supplies of oil and gas are limited. The main offshore oil fields are in the North Sea, Gulf of Mexico, Persian Gulf, and along the coasts of South America and Asia.



WIND TURBINES
Unlike oil, wind turbines are a source of renewable energy. Some thirty turbines operate at North Hoyle Offshore Wind Farm, located 6 miles (10 km) off Rhyl in North Wales, UK.



DEATH AND DECAY
Plant and bacteria remains from ancient seas fell to the sea floor and were covered by mud layers. Heat and pressure turned them into oil, then gas, which moved up through porous rocks, to be trapped by impermeable rocks.

AT WORK
On a drilling platform, workers are looking after the drill bit that is used to cut down through the rocks on the seabed. When in operation, special mud is sent down pipes attached to the drill bit to cool it, wash out the ground-up rock and prevent the oil from gushing out.

ON FIRE
Oil and gas are highly flammable. Despite precautions, accidents do happen, like the North Sea's *Piper Alpha* disaster in 1988 when 167 people died. Since then safety measures have been improved.

Tallest structure on this platform is flare stack for safety reasons

FLOATING PRODUCTION
The oil industry increasingly uses floating vessels called FPSOs (Floating Production, Storage, Offloading). These vessels take the oil and gas produced from nearby drilling platforms and undersea wells. The FPSO processes the oil and stores it, until it is taken away by tankers. FPSOs work well in deepwater locations which are too far to connect to the shore by seabed pipelines. The vessels can also move out of the way of hurricanes or drifting icebergs. They can also move when an oil field is exhausted.